Prisoner's Lemma: Exploitability-Aware Reinforcement Learning for Online Strategic Adaptation

Stanford CS224R Final Project

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1 Extended Abstract

Reinforcement learning agents trained to maximize average return can learn policies that perform well against a training distribution while remaining strategically vulnerable against particular opponent regimes. We study this problem in the Iterated Prisoner's Dilemma (IPD), a minimal repeated game that isolates the core tension between exploitation and reciprocal cooperation. IPD is particularly useful as a testbed because the one-shot dominant strategy of defection is well-known, yet conditionally cooperative strategies such as Tit-for-Tat outperform it under iterated play against a broad opponent population, making the gap between average-return optimization and strategic robustness diagnosable and interpretable.

The project began with vanilla REINFORCE baselines. Self-play and fixed opponent-population training both repeatedly collapsed to unconditional defection. Learned policies achieved high return against Always Cooperate, Always Defect, Random, and Win-Stay Lose-Shift, but only about 104 return against Tit-for-Tat and Grim Trigger over horizon 100. Sweeps over history length, discount, entropy, and reward normalization changed which unconditional attractor was reached, but did not produce robust conditional adaptation. However, training on a clean reciprocal core of Tit-for-Tat and Grim Trigger achieved normalized score 1.000, showing that cooperation itself was learnable. The harder problem was learning when to cooperate versus defect.

We therefore implemented exploitability-aware gap weighting (EA Gap). For each opponent regime, the agent tracks a smoothed exponential moving average of its gap to a best-known response return, and weights training updates toward opponents where this gap remains large. Unlike PSRO or NFSP, EA Gap requires no policy population, no response oracle, and no opponent label at deployment. It is a lightweight harness on top of standard REINFORCE that makes opponent-specific strategic failures harder to ignore during training without changing the policy architecture or observation space.

In the final hidden-opponent suite, policies were evaluated against six fixed opponent regimes with no opponent identity provided at deployment. Vanilla contextual RL barely improved over no-context RL, increasing broad score from 0.743 to 0.751. EA Gap improved the same contextual shared policy to 0.815, increasing mean normalized return from 0.839 to 0.899, worst-opponent return from 0.616 to 0.675, and success fraction from 0.700 to 0.783. The gain was opponent-specific: Tit-for-Tat and Grim Trigger improved from 0.616 to 0.880, while Always Cooperate decreased from 0.840 to 0.675. Thus, EA Gap changed strategic posture toward reciprocal robustness rather than uniformly improving every matchup. As an oracle upper bound, opponent-label-routed experts reached score 0.894, confirming that strong conditional strategies are representable when regime selection is solved. Latent modular policies with end-to-end learned routers underperformed shared EA Gap even with gap weighting, suggesting that routing and expert specialization do not emerge reliably from reward alone.

We also tested whether the bottleneck was inference or policy capacity. A supervised probe classified opponent regime from four-episode behavioral summaries with 0.791 accuracy compared to chance 0.167, showing that sufficient opponent-regime information exists in context. The gap between this probe accuracy and vanilla RL's near-zero benefit from context (0.743 to 0.751) defines an inference-control gap: the information is present but the policy does not reliably use it. These results suggest that robust strategic adaptation requires not only learning useful behaviors, but also developing mechanisms to select the right behavior from interaction history. EA Gap narrows this gap through training pressure rather than architectural change, but fully solving the inference-control problem will likely require stronger representations, auxiliary opponent-prediction objectives, or belief-state learning.

2 Abstract

We study exploitability-aware reinforcement learning in the Iterated Prisoner's Dilemma, where naive reward maximization collapses to unconditional defection despite diverse opponent training. We introduce EA Gap, a lightweight gap-weighting objective that increases training pressure on opponent regimes where the agent's return most lags a best-known response target, without providing opponent labels at deployment. In hidden-opponent evaluations against six fixed strategies, EA Gap improves broad strategic score from 0.751 to 0.815 over a vanilla contextual baseline, with the largest gains against reciprocal and retaliatory opponents. A behavioral probe shows that sufficient opponent-regime information exists in context but is not exploited by vanilla RL, framing robust strategic adaptation as an inference-control problem.

3 Introduction

Reinforcement learning is typically formulated as the maximization of expected return under an environment distribution. Vanilla RL through REINFORCE is best suited to stationary objectives, but in multi-agent settings the effective objective changes with the opponent's strategy, making a single globally optimal behavior difficult to define. Furthermore, a policy can perform well on average while remaining highly exploitable by a particular opponent type, creating a mismatch between average return and strategic robustness that appears even in a small repeated game such as iterated prisoner's dilemma (IPD). Defecting against a naive cooperator gives high immediate reward, but defecting against a retaliatory opponent can destroy future cooperation. Thus, a policy that learns broad defection may look strong against a mixed population, while failing the key strategic test of preserving cooperation with reciprocal opponents. Conversely, a policy that learns broad cooperation may be beneficial against reciprocal opponents but leave reward on the table against naive opponents. The same exploitability tradeoff appears in real-world multi-agent settings such as automated negotiation, algorithmic trading, and autonomous vehicle interaction, where short-run gains can change how other agents respond over time.

We are interested in studying game theory-inspired methods to improve broad online performance of RL agents against a range of opponent strategy profiles in an IPD environment, testing specifically the hypothesis:

Exploitability-aware training can help an RL agent to reduce its strategic vulnerability in an IPD scenario across a broad range of hidden opponent profiles.

4 Related Work

Multi-agent reinforcement learning (MARL) has long recognized that naive reward maximization can fail in strategic settings. Self-play can produce strong agents, but may cycle, overfit to its own distribution, or converge to policies that exploit training regularities rather than learning robust strategic behavior.

Fictitious play and Neural Fictitious Self-Play (NFSP) aim to approximate equilibrium behavior by learning best responses to average opponent strategies. Policy-Space Response Oracles (PSRO) extend this idea through policy populations and iterative best-response training. These methods explicitly incorporate game-theoretic structure, but require maintaining policy populations or response oracles. This project further asks what can be learned from a lightweight exploitability-aware harness in a controlled hidden-opponent setting.

Large-scale systems such as DeepNash demonstrate that game-theoretic objectives and regularized dynamics can produce strong performance in complex imperfect information games. CICERO shows the importance of opponent modeling and strategic reasoning in the environment of a large action-space online strategy game 'Diplomacy'. These systems are important motivation, but they use substantial domain-specific machinery. Our goal is to use IPD as a microscope for isolating when simple RL agents fail to use opponent information and whether exploitability-aware pressure can reduce that failure, where IPD was chosen as the environment due to its small, diagnosable action-space, and further the information-rich contrast of a known maximizing strategy of defecting in any one-shot Prisoner's Dilemma versus superior reward maximization from conditionally cooperative strategies for many opponent profiles when the same game is repeated many times against the same opponent.

This work also connects to opponent modeling and latent-strategy inference. In large strategic environments, the agent may face latent opponent styles, including aggressive rushers, defensive players, opportunists, expansionists, and retaliatory opponents. The agent should not need an explicit label for these styles, but should infer them from behavior. IPD offers a minimal version of this problem, where latent opponent regimes are fixed and interpretable.

5 Method

All learned agents use policy-gradient training from opponent interaction in-situ. The baseline is a shared REINFORCE policy that samples opponents from the training population and optimizes observed return. In the deployable hidden-opponent setting, the policy is never given the opponent label, and its in-episode observation is the flattened joint-action history. In `ctx4` runs this is concatenated with an eight-dimensional summary of the previous four episodes against the same hidden opponent, including cooperation rates, mutual cooperation, normalized return, opponent response after agent cooperation or defection, and first-defection timing.

For reference throughout this work, best-known response return refers to the best return achieved by our fixed-strategy benchmark pool against a given opponent. It serves as an empirical normalization target, and not a formal proof of optimal play.

The exploitability-aware variant uses the same policy class and context, but changes training pressure. For each opponent o, we compute normalized return relative to a best-known response target and track a smoothed opponent-specific gap,

$$\mathrm{Gap}(o) = \max(0, 1 - \mathrm{NormReturn}(o)).$$

A large gap means the agent is consistently underexploiting that opponent compared to the best-known response. During gap-weighted training, updates from opponent o are weighted by $1 + \lambda \mathrm{Gap}_{\mathrm{EMA}}(o)$. This does not tell the agent the correct action or provide an opponent label at evaluation, but makes failures that average return would hide harder to ignore during training.

Algorithm 1 EA Gap training loop

Initialize policy π_θ and $\mathrm{Gap}_{\mathrm{EMA}}(o) \leftarrow 1$ for each opponent o
for each update **do**
  for each opponent o **do**
    Collect one adaptation block against o
    Compute $\mathrm{NormReturn}(o)$ for the block
    $g(o) \leftarrow \max(0, 1 - \mathrm{NormReturn}(o))$
    $\mathrm{Gap}_{\mathrm{EMA}}(o) \leftarrow \beta \mathrm{Gap}_{\mathrm{EMA}}(o) + (1 - \beta) g(o)$
    $w(o) \leftarrow 1 + \lambda \mathrm{Gap}_{\mathrm{EMA}}(o)$
  end for
  Apply a REINFORCE update with each opponent block weighted by $w(o)$
end for

We also test a latent modular policy implemented as a soft mixture of experts. The policy first maps the observation and context into a shared hidden state. Four expert heads each map this hidden state to an action distribution, while a router head maps the same hidden state to a softmax distribution over experts. The final action distribution is the router-weighted mixture of the four expert action distributions. The router receives the same observation and context as the shared policy, but no opponent label. Experts and router are trained jointly by the same REINFORCE loss, with no expert pretraining, opponent-classification loss, hard expert assignment, or explicit diversity regularization.

Next, we test opponent-label-routed experts as a diagnostic upper bound, though not deployable in the established online IPD setting. These experts, who are given the strategy regime of each opponent they play against, serve to test whether strong conditional behavior can arise from a modular approach if regime selection is solved.

Returns are normalized by best-known response performance. A normalized return of 1.0 indicates that the agent matches the best-known response level against that opponent. We report mean normalized return, worst-opponent normalized return, and success fraction, where 'success' is defined as a normalized return at least 0.95. These metrics are combined into a broad strategic score:

$$\text{Score} = 0.55 \cdot \text{Mean} + 0.30 \cdot \text{Worst} + 0.15 \cdot \text{Success}.$$

This score is an evaluation metric that emphasizes average performance while still penalizing policies that fail badly against any single opponent regime.

6 Experimental Setup

The main testbed is iterated prisoner's dilemma with payoff matrix:

	Opponent Cooperates	Opponent Defects
Agent Cooperates	3	0
Agent Defects	5	1

Thus, mutual cooperation gives stable medium reward, exploiting a cooperator gives high immediate reward, mutual defection gives low reward, and being exploited gives zero reward. Final hidden-opponent experiments use horizon 50.

We evaluate against six fixed opponent regimes:

- **Always Cooperate**: always cooperates.
- **Always Defect**: always defects.
- **Random**: cooperates or defects stochastically.
- **Tit-for-Tat**: starts by cooperating, then copies the agent's previous action.
- **Grim Trigger**: starts by cooperating, but defects forever after the agent defects once.
- **Win-Stay Lose-Shift**: starts by cooperating, repeats its previous action after payoff at least 3, and switches after lower payoff.

These opponents expose distinct strategic pressures: Always Cooperate rewards exploitation, Always Defect and Random reward defection, Tit-for-Tat and Grim Trigger reward sustained cooperation and punish early defection, and Win-Stay Lose-Shift introduces payoff-conditioned adaptation. A key challenge is thus conditional behavior, in that an optimal agent strategy should defect against exploitable or hostile opponents but cooperate with reciprocal and retaliatory opponents. REINFORCE was chosen as the baseline because it is a standard policy-gradient method with no opponent-modeling assumptions, and the $0.55/0.30/0.15$ score weights emphasize average performance while still penalizing worst-case failure.

Final hidden-opponent runs used 3000 policy-gradient updates, horizon $H = 50$, history length 20, discount $\gamma = 0.99$, learning rate 10^{-2}, hidden dimension 64, entropy coefficient 0.05, four-episode context for `ctx4`, 50 evaluation blocks per opponent, EA gap weight strength $\lambda = 4.0$, gap EMA decay 0.95, and four expert heads for latent modular policies. Each final update used four-episode adaptation blocks. For balanced and EA Gap training, one block was collected for each opponent per update, while vanilla sampled training used the same number of blocks sampled uniformly from the opponent pool.

7 Results

7.1 Quantitative Evaluation

7.1.1 Main hidden-opponent comparison

Table 1 reports the final hidden-opponent online results. All rows use 10 seeds. The primary deployable comparison is Vanilla `ctx4`versus EA Gap `ctx4`, where both use the same hidden-opponent setup and context, but differ in exploitability-aware training pressure.

EA Gap improves broad score from 0.751 to 0.815. It also improves mean normalized return, worst-opponent return, and success fraction. This supports the central claim that exploitability-aware gap weighting improves the clean online baseline.

Method	Score	Mean	Worst	Success
Vanilla `ctx0`	0.743	0.828	0.624	0.667
Vanilla `ctx4`	0.751	0.839	0.616	0.700
EA Gap `ctx4`	0.815	0.899	0.675	0.783
Latent `ctx4`	0.675	0.789	0.488	0.633
Latent EA `ctx4`	0.728	0.806	0.616	0.667

Table 1: Final hidden-opponent online results. All rows use 10 seeds.

7.1.2 Opponent-specific effects

The improvement is not uniform, as EA Gap primarily improves reciprocal and retaliatory opponents. Tit-for-Tat and Grim Trigger normalized returns increase from 0.616 to 0.880, and Always Cooperate decreases from 0.840 to 0.675. We see that this shift driven by reducing exploitability by playing 'safer' strategies does reduce exploitability but simultaneously worsens the policy's ability to exploit naive exploitable opponents.

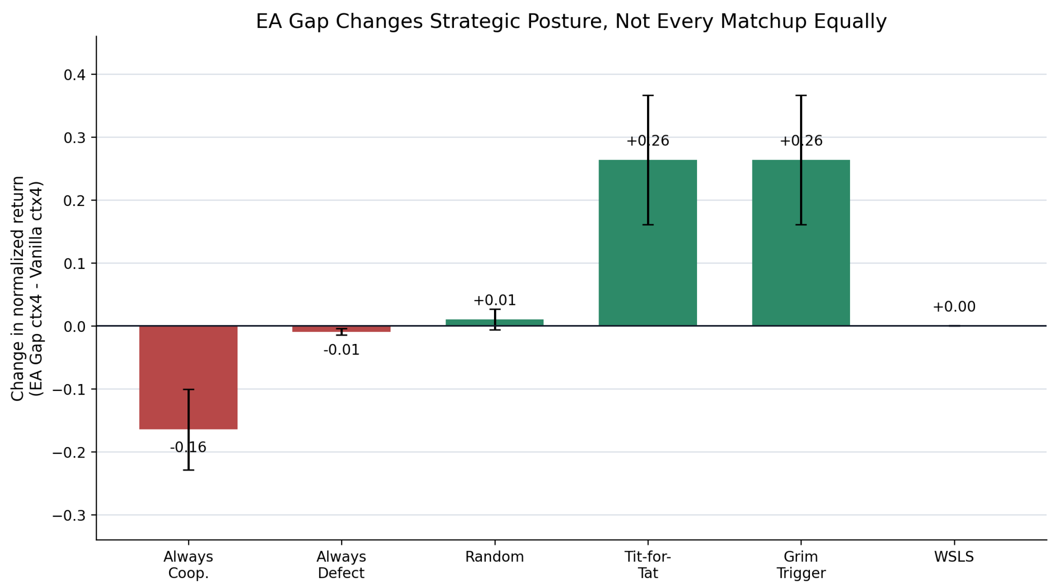

Figure 1: Change in normalized return from Vanilla `ctx4`to EA Gap `ctx4`by opponent. EA Gap improves Tit-for-Tat and Grim Trigger while reducing Always Cooperate exploitation.

7.1.3 Opponent-Regime Probe

To test whether context actually contains opponent-regime information, we trained a lightweight classifier to predict the hidden opponent from the same style of behavioral summaries available to the policy. We use a multinomial logistic-regression classifier trained on standardized context-summary features using a $70/30$ train-test split, 1200 samples per opponent, five seeds, learning rate 0.25, 700 epochs, and 10^{-4} L2 regularization. This classifier is not part of the RL agent, but rather a diagnostic probe.

The probe shows that four episodes of behavioral context make opponent regimes statistically readable. However, vanilla RL barely benefits from context alone, as Vanilla `ctx0`scores 0.743, while Vanilla `ctx4`scores 0.751. Thus, we see an 'inference-control gap' where necessary information for opponent regime identification is present, but vanilla RL does not reliably use it to choose the correct strategic response.

Context Window	Opponent ID Accuracy
`ctx0`	0.167 ± 0.000
`ctx1`	0.614 ± 0.007
`ctx2`	0.722 ± 0.013
`ctx4`	0.791 ± 0.009

Table 2: Opponent-regime classification accuracy from context summaries. Chance accuracy is $1/6 = 0.167$.

This result also helps interpret latent modularity's performance, by indicating that the failure is less likely to be that no information is available, and more likely to involve the difficulty of learning routing and expert specialization through reward alone.

7.2 Qualitative Analysis

7.2.1 Chronological Diagnostic Experiments

Here we present the results from numerous experiments shaping design choices in the final set of hidden-opponent experiments.

1. Vanilla self-play collapses to mutual defection. We began with two independent REINFORCE agents trained in self-play for 5000 episodes across three seeds. All three converged to mutual defection, where final player returns were $100/100$ over horizon 100, cooperation rates were 0, and mutual cooperation was 0. When the learned player-0 policy (arbitrarily the first agent from each seed's two learned agents) was frozen and evaluated against the fixed opponent pool, it behaved like Always Defect, with return 500 against Always Cooperate, 100 against Always Defect, about 301 against Random, 300 against WSLS, and only 104 against Tit-for-Tat and Grim Trigger. This established the first failure mode, in that self-play collapsed to the one-shot-optimal Prisoner's Dilemma equilibrium strategy of playing Always Defect regardless of opponent behavior.

2. Fixed-opponent population training does not fix the attractor. We next trained a single REINFORCE policy against a randomly sampled fixed opponent from the six-opponent pool in each episode. This policy collapsed to all-defect in all three seeds, same as the previous self-play result. Final fixed-opponent returns were essentially 500 against Always Cooperate, 100 against Always Defect, about 299 against Random, 300 against WSLS, and 104 against Tit-for-Tat and Grim Trigger, with evaluation cooperation rate 0. This showed that exposure to a diverse opponent population is not enough; because four of the six regimes do not strongly punish it, average return rewards broad defection.

3. Fixed-strategy oracles reveal what average return hides. We evaluated fixed strategies against the same opponent pool of fixed strategies to define best-known response targets. Conditional strategies were strongest on average. Grim Trigger achieved mean return 266.19, Tit-for-Tat 253.78, WSLS 245.60, Always Defect 234.84, Always Cooperate 225.10, and Random 197.82. We see from these results that while Always Defect has large opponent-specific value gaps against reciprocal opponents, it appears strong under mean return with a relatively high overall score, thus demonstrating a local minimum attractor in Always Defect and explaining the collapse to an opponent-agnostic Always Defect policy when training Vanilla RL to maximize overall reward.

4. Memory, discount, entropy, and reward scaling are insufficient. We then swept population RL variants over history length, discount, and entropy. The best broad scores in this sweep were only around 0.67, and the policies still fell into unconditional modes rather than clean opponent-adaptive behavior. Best-known-response-normalized reward training also converged to all-defect in all three seeds, where Tit-for-Tat and Grim Trigger remained at return 104, giving a best-known-response gap of 196 at horizon 100, while Always Cooperate, Always Defect, Random, and WSLS remained near their best-known-response targets. This ruled out insufficient memory, exploration, or reward-scale normalization as the key obstacles toward optimal, conditional reward maximization strategies.

Figure 2: Fixed-strategy returns used to define best-known response targets. We see different opponents require incompatible responses.

5. Clean reciprocity is learnable, but mixed curricula break it. To isolate whether the learner could discover cooperation at all, we trained on the clean reciprocal core containing only Tit-for-Tat and Grim Trigger. Across three seeds, mean normalized return, worst normalized return, and success fraction were all 1.000. However, curriculum variants revealed that this competence did not automatically survive new opponent type. Raw reciprocal core plus Always Defect still performed well after longer training, with mean normalized return 0.969, worst 0.907, and perfect Tit-for-Tat/Grim performance, but broader gated curricula averaged only about 0.665 mean normalized return and 0.160 worst-opponent normalized return. Figure 3 summarizes the curriculum diagnostics: reciprocal cooperation is learnable, but conditional switching is fragile.

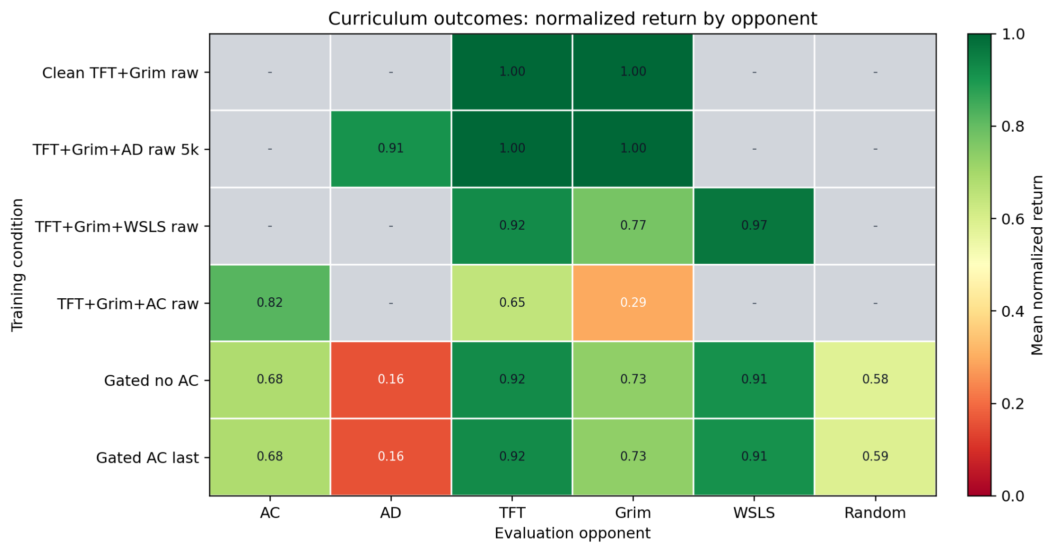

Figure 3: Curriculum diagnostics. Blank cells are opponents not included in that run. The experiments show that reciprocal cooperation can be learned in clean settings, but adding exploitation-rich opponents can pull the policy back toward broad global modes.

6. Exploitability diagnostics motivated the final EA Gap objective. The curriculum and best-known-response-gap diagnostics showed that average return could hide large opponent-specific failures, especially against Tit-for-Tat and Grim Trigger. This motivated the final EA Gap formulation, designed to track smoothed opponent-specific gaps to best-known response performance and use those gaps to weight training pressure toward regimes where the policy remains vulnerable.

7. Opponent-label experts separate representation from inference. Finally, we trained opponent-label-routed experts, where experts were provided with a label indicating the fixed opponent type in each episode, as an oracle diagnostic. This condition reached IPD score 0.894, higher than the deployable hidden-opponent methods. This showed that strong conditional strategies are representable when regime selection is solved. The remaining question therefore became, when the opponent label is hidden, can behavioral context and exploitability-aware pressure help a shared policy reduce strategic vulnerability?

7.2.2 Behavior versus quality

For example, against Tit-for-Tat, an EA Gap policy often opens with cooperation and then continues producing long runs of mutual cooperation rather than triggering retaliation. Against Always Cooperate, the same policy can defect early and remain mostly defecting, illustrating that the learned behavior is not simply higher cooperation everywhere.

Figure 4 compares cooperation rate with normalized return. Notably, EA Gap changes behavior in strategically specific ways rather than simply increasing or decreasing cooperation everywhere.

Figure 4: Learned strategic behavior by opponent. Left: cooperation rate. Right: normalized return, where 1.0 matches the best-known response level.

7.2.3 Seed-level strategic modes

Across seeds, policies mostly converge to broad defection or reciprocal safety. We classify broad defection as strong Always Cooperate/Always Defect performance (≥ 0.80 norm) but weak Tit-for-Tat/Grim performance (< 0.70 average norm). We classify reciprocal safety as strong Tit-for-Tat/Grim performance (≥ 0.80 average norm and ≥ 0.75 average cooperation) but weak Always Cooperate exploitation (< 0.90 norm). Conditional near-optimal additionally requires low cooperation and high return against Always Cooperate, Always Defect, and Random, plus WSLS norm ≥ 0.90.

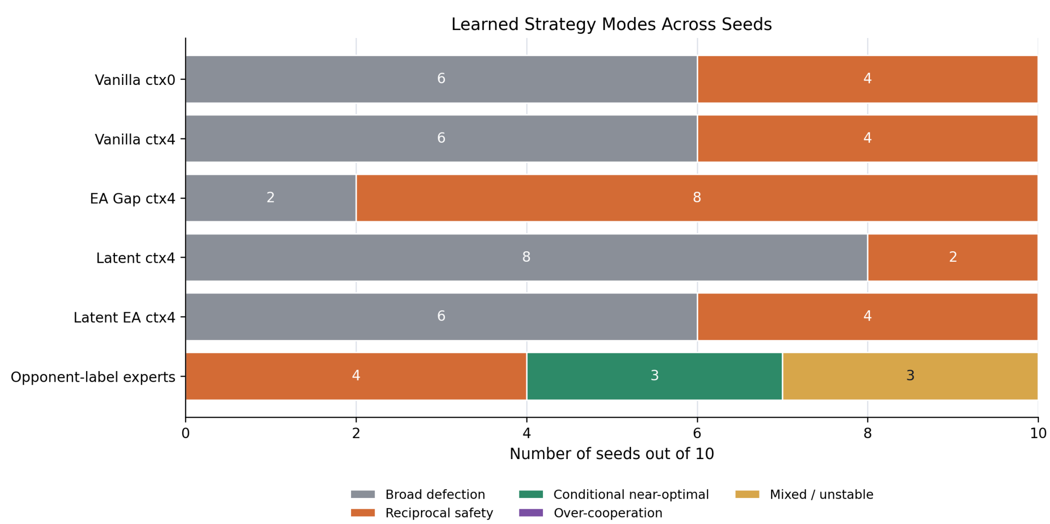

Figure 5: Seed-level strategic modes across methods. Vanilla contextual RL most often converges to broad defection, while EA Gap shifts more seeds toward reciprocal safety. Latent modular policies do not reliably produce clean conditional specialization without opponent labels or auxiliary routing supervision.

RL convergence is stochastic, and the result shows that the training objective changes the distribution over convergence basins. EA Gap reduces broad-defection outcomes from 6/10 seeds to 2/10 seeds and increases reciprocal-safety outcomes from 4/10 to 8/10. However, it does not produce fully conditional best-response behavior.

8 Discussion

The experiments support a nuanced version of the original, where exploitability-aware training can improve hidden-opponent strategic robustness, but it does not universally solve strategic convergence. The strongest positive finding is that EA Gap improves the clean hidden-opponent contextual baseline, while the strongest negative finding is that information availability is not enough. Regarding the latter, we find that context does contain opponent-regime signal, but vanilla RL barely benefits from it.

Seed-by-seed analysis demonstrates learned policies tend to fall into two simple attractors. Broad defection is profitable against Always Cooperate, Always Defect, Random, and often WSLS, but fails against Tit-for-Tat and Grim Trigger. Reciprocal safety preserves cooperation with reciprocal opponents but sacrifices exploitation of naive cooperators. Conditional best response requires selecting between these behaviors based on opponent regime. Opponent-label experts show this is possible when regime selection is solved, but hidden-opponent policies do not reliably learn it.

Latent modularity was not sufficient. Latent EA improves over latent vanilla, but remains below shared EA. This suggests that simply adding experts is not enough, and that the training process must also make experts specialize and make the router select them reliably. Future latent approaches may need auxiliary opponent-inference losses, expert-diversity regularization, belief-state learning, or supervised warm starts.

This is a controlled study in a small environment. IPD has binary actions, fixed opponent regimes, and hand-specified best-known response estimates. The results do not directly establish performance in large strategic games. While the external and supervised opponent-regime probe shows that information is recoverable from the context provided, the latent router was not directly evaluated for routing accuracy, so we cannot conclusively attribute latent failure to classification error.

The final hidden-opponent comparisons use 10 seeds per condition, such that while effects of different training methodologies are meaningful, they should be interpreted as evidence that EA Gap shifts convergence tendencies, rather than as proof of guaranteed convergence to optimal play. Our broad score metric as earlier defined encodes a robustness preference, whereas different applications might value average reward, worst-case safety, or exploitative opportunity differently.

9 Future Work

The most important next direction is improving the inference-control interface. Since context contains opponent-regime information, future agents should learn representations that make that information usable. Possible approaches include auxiliary opponent prediction, contrastive learning over opponent behavior, recurrent belief states, or explicit latent-style inference.

Second, modular policies should be revisited with stronger specialization pressure. Opponent-label experts show that conditional experts can work, but end-to-end latent routing did not discover reliable specialization. Expert diversity regularization, router entropy control, or supervised pretraining of routers may help.

Third, exploitability estimation should be made less hand-specified. Rather than using best-known response tables, future work could use learned response oracles, policy populations, or regret estimates. This would connect the lightweight EA Gap idea more directly to PSRO-style training while preserving a focus on online adaptation.

Finally, the framework should be tested in larger strategic microgames before full-scale environments. Useful intermediate settings include resource races, territory control, tactical combat, expansion timing, and diplomacy-like coordination/defection. The guiding question should remain, can the agent infer latent strategic style from behavior and use that inference to reduce exploitability?

10 Conclusion

We began by asking whether exploitability-aware RL could improve convergence to game-theoretically robust strategies. The answer is partially but not wholly yes. Naive average-reward RL often converges to broad defection. We find that clean reciprocal cooperation is learnable, and opponent-label experts show that strong conditional strategies exist. In the hidden-opponent setting, behavioral context contains enough information to identify opponent regimes, but vanilla RL does not reliably use it. Exploitability-aware gap weighting improves the shared contextual policy, shifting behavior toward reciprocal robustness and improving broad strategic score from 0.751 to 0.815.

The primary takeaway is that game-theoretically-optimizing RL in IPD is an inference-control problem. Agents must learn good behaviors, but also infer when each behavior applies. Exploitability-aware training helps expose and reduce strategic vulnerabilities, but robust online strategic adaptation will require stronger mechanisms for opponent inference, regime selection, and conditional control.

11 AI Tools Disclosure

GPT 5.5 was used only for writing boilerplate code, debugging, and implementing the code for simple scripts to produce plots for data representation.

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